Problem Set # 11

S 8.6.2

$$L = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} m \omega^2 x^2$$

$$\ddot{x} = -\omega^2 x$$

$$x(t) = A \sin(\omega t + \delta)$$

$$\begin{cases} x(t') = A \sin(\omega t' + \delta) = x' \\ x(t'') = A \sin(\omega t'' + \delta) = x'' \end{cases}$$

Let us choose $t' = 0$

$$\begin{cases} A \sin\delta = x' \\ A(\sin\omega t'' \cos\delta + \cos\omega t'' \sin\delta) = x'' \end{cases}$$

$$x(t) = A \sin\omega t \cos\delta + A \cos\omega t \sin\delta$$

$$= \frac{\sin\omega t}{\sin\omega t''}(x'' - \cos\omega t'' \, x') + \cos\omega t \, x'$$

$$S_{cl} = \int_0^{t''} dt \; \frac{1}{2} m A^2 \omega^2 [\cos^2(\omega t + \delta) - \sin^2(\omega t + \delta)]$$

$$= \frac{1}{2} m\omega^2 A^2 \int_0^{t''} dt \; \cos[2(\omega t + \delta)]$$

$$S_{cl} = \frac{1}{2} m \omega^2 A^2 \frac{1}{2\omega} \left[\sin\left[2(\omega t'' + \delta)\right] - \sin 2\delta \right]$$

$$= \frac{1}{2} m \omega A^2 \left[\sin(\omega t'' + \delta) \cos(\omega t'' + \delta) - \sin\delta \cos\delta \right]$$

$$= \frac{1}{2} m \omega \left\{ x'' \left[\frac{\cos\omega t''}{\sin\omega t''} (x'' - \cos\omega t'' x') - \sin\omega t'' x' \right] \right.$$

$$\left. - x' \frac{1}{\sin\omega t''} (x'' - \cos\omega t'' x') \right\}$$

$$= \frac{1}{2} \frac{m\omega}{\sin\omega t''} \left[\cos\omega t'' (x''^2 + x'^2) - 2 x' x'' \right]$$

Therefore

$$U(x, t; x') = A(t) \exp\left\{ \frac{i m \omega}{2\hbar \sin\omega t} \left[(x^2 + x'^2) \cos\omega t - 2 x x' \right] \right\}$$

8.6.3 $\quad U(x,t;x',t') = \sum_n \psi_n(x)^* \psi_n(x')\, e^{-\frac{i}{\hbar}E_n(t-t')}$

For the harmonic oscillator (see section 8.6 and exercise 8.6.2)

$$U(x,t;x',t') = A(t)\, e^{\frac{im\omega}{2\hbar \sin\omega t}\left[(x^2+x'^2)\cos\omega t - 2xx'\right]}$$

i. We are told $\quad A(t) = \sqrt{\dfrac{m\omega}{2\pi i\hbar \sin\omega t}}$

$$U(0,t;0,0) = \sum_n |\psi_n(0)|^2 e^{-\frac{i}{\hbar}E_n t}$$

$$= \sqrt{\frac{m\omega}{2\pi i\hbar \sin\omega t}}$$

Expand in powers of $\alpha' = \frac{1}{\alpha} = e^{-i\omega t}$

$$U(0,t;0,0) = \sqrt{\frac{m\omega}{\pi\hbar}} \frac{1}{\left(\frac{1}{\alpha'} - \alpha'\right)^{1/2}}$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}} \sqrt{\alpha'} \frac{1}{(1-\alpha'^2)^{1/2}}$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}} \sqrt{\alpha'} \left[1 + \sum_{n=1}^{\infty} \frac{1}{n!} (-\alpha'^2)^n \left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)\cdots\left(\frac{1}{2}-n\right)\right]$$

$$= \frac{1}{\sqrt{\pi}\,\Delta} \left[e^{-\frac{i}{2}\omega t} + \frac{1}{2} e^{-\frac{5i}{2}\omega t} + \frac{3}{8} e^{-\frac{9i}{2}\omega t} + \cdots \right]$$

$$\left(\Delta \equiv \sqrt{\frac{\hbar}{m\omega}}\right)$$

Only the $E_n = \hbar\omega\left(\frac{1}{2} + n\right)$ with $n = 0, 2, 4 \ldots$ appear because

$$\psi_n(0) = 0 \quad \text{for } n \text{ odd}$$

Check:

$$|\psi_0(0)|^2 = \frac{1}{\sqrt{\pi}\,\Delta} (H_0(0))^2 = \frac{1}{\sqrt{\pi}\,\Delta} \quad \checkmark$$

$$|\psi_2(0)|^2 = \frac{1}{2^2\, 2!\, \sqrt{\pi}\,\Delta}\left(H_2(0)\right)^2 = \frac{1}{2}\,\frac{1}{\sqrt{\pi}\,\Delta}$$ ✓

$$|\psi_4(0)|^2 = \frac{}{2^4\, 4!\, \sqrt{\pi}\,\Delta}\left(H_4(0)\right)^2 = \frac{(12)^2}{2^4\, 24\, \sqrt{\pi}\,\Delta}$$

$$= \frac{3}{8}\,\frac{1}{\sqrt{\pi}\,\Delta}$$ ✓

<u>ii</u>.

$$U(x,t;x,0) = \sum_n |\psi_n(x)|^2\, e^{-\frac{i}{\hbar}E_n t}$$

$$= \sqrt{\frac{m\omega}{2\pi i\hbar \sin\omega t}}\; e^{i\frac{m\omega}{\hbar \sin\omega t}x^2(\cos\omega t - 1)}$$

Epand in powers of $\alpha' = \frac{1}{\alpha} = e^{-i\omega t}$

$$U(x,t;x,0) = \sqrt{\frac{m\omega}{\pi\hbar}}\;\frac{1}{\left(\frac{1}{\alpha'}-\alpha'\right)^{1/2}}\; e^{-\frac{m\omega x^2}{\hbar}\left(\frac{1+\alpha'^2-2\alpha'}{1-\alpha'^2}\right)}$$

$$= \frac{\sqrt{\alpha'}}{\sqrt{\pi}\,\Delta}\left(1+O(\alpha'^2)\right)\left[e^{-\frac{x^2}{\Delta^2}} + \frac{2x^2}{\Delta^2}\alpha'\, e^{-\frac{x^2}{\Delta^2}} + O(\alpha'^2)\right]$$

$$E_0 = \frac{\hbar\omega}{2} \qquad |\psi_0(x)|^2 = \frac{1}{\sqrt{\pi}\,\Delta} e^{-\frac{x^2}{\Delta}}$$

$$E_1 = \frac{3\hbar\omega}{2} \qquad |\psi_1(x)|^2 = \frac{2x^2}{\sqrt{\pi}\,\Delta^3} e^{-\frac{x^2}{\Delta^2}}$$

S10.1.2

a.

$$\sigma_1^{(1)\otimes(2)} \begin{pmatrix} |+\rangle\otimes|+\rangle \\ |+\rangle\otimes|-\rangle \\ |-\rangle\otimes|+\rangle \\ |-\rangle\otimes|-\rangle \end{pmatrix} = \begin{pmatrix} (a|+\rangle + b|-\rangle)\otimes|+\rangle \\ (a|+\rangle + b|-\rangle)\otimes|-\rangle \\ (c|+\rangle + d|-\rangle)\otimes|+\rangle \\ (c|+\rangle + d|-\rangle)\otimes|-\rangle \end{pmatrix}$$

$$= \begin{pmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{pmatrix} \begin{pmatrix} |+\rangle\otimes|+\rangle \\ |+\rangle\otimes|-\rangle \\ |-\rangle\otimes|+\rangle \\ |-\rangle\otimes|-\rangle \end{pmatrix}$$

b.

$$\sigma_2^{(1)\otimes(2)} \begin{pmatrix} |+\rangle\otimes|+\rangle \\ |+\rangle\otimes|-\rangle \\ |-\rangle\otimes|+\rangle \\ |-\rangle\otimes|-\rangle \end{pmatrix} = \begin{pmatrix} |+\rangle\otimes(e|+\rangle + f|-\rangle) \\ |+\rangle\otimes(g|+\rangle + h|-\rangle) \\ |-\rangle\otimes(e|+\rangle + f|-\rangle) \\ |-\rangle\otimes(g|+\rangle + h|-\rangle) \end{pmatrix}$$

$$= \begin{pmatrix} e & f & 0 & 0 \\ g & h & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{pmatrix} \begin{pmatrix} |+\rangle\otimes|+\rangle \\ |+\rangle\otimes|-\rangle \\ |-\rangle\otimes|+\rangle \\ |-\rangle\otimes|-\rangle \end{pmatrix}$$

c.

$$\left[(\sigma_1\sigma_2)^{(1)\otimes(2)}\right]_{(ij)(kl)} = (\sigma_1)_{ik}(\sigma_2)_{jl}$$

$$\therefore\ (\sigma_1\sigma_2)^{(1)\otimes(2)} = \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix}$$

$$\sigma_1^{(1)} \otimes \sigma_2^{(2)} \begin{pmatrix} |+\rangle \otimes |+\rangle \\ |+\rangle \otimes |-\rangle \\ |-\rangle \otimes |+\rangle \\ |-\rangle \otimes |-\rangle \end{pmatrix} = \begin{vmatrix} (a|+\rangle + b|-\rangle) \otimes (e|+\rangle + f|-\rangle) \\ (a|+\rangle + b|-\rangle) \otimes (g|+\rangle + h|-\rangle) \\ (c|+\rangle + d|-\rangle) \otimes (e|+\rangle + f|-\rangle) \\ (c|+\rangle + d|-\rangle) \otimes (g|+\rangle + h|-\rangle) \end{vmatrix}$$

$$= \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cd & de & df \\ cg & ch & dg & dh \end{pmatrix} \begin{pmatrix} |+\rangle \otimes |+\rangle \\ |+\rangle \otimes |-\rangle \\ |-\rangle \otimes |+\rangle \\ |-\rangle \otimes |-\rangle \end{pmatrix}$$

2

S10.2.3

$$H = \frac{P_x^2 + P_y^2 + P_z^2}{2m} + \frac{m\omega^2}{2}(x^2 + y^2 + z^2)$$

i. $$H = H_x + H_y + H_z$$

$$H_x = \frac{1}{2m} P_x^2 + \frac{m\omega^2}{2} x^2 \quad , \quad \text{etc..}$$

$$H | E_{n_x n_y n_z} \rangle = E_{n_x n_y n_z} | E_{n_x n_y n_z} \rangle$$

$$n_x, n_y, n_z = 0, 1, 2 \ldots$$

$$H_x | E_{n_x} \rangle = E_{n_x} | E_{n_x} \rangle \quad , \text{etc..}$$

$$E_{n_x n_y n_z} = E_{n_x} + E_{n_y} + E_{n_z}$$

$$= \hbar\omega \left(n_x + \tfrac{1}{2} + n_y + \tfrac{1}{2} + n_z + \tfrac{1}{2}\right)$$

$$= \hbar\omega \left(n + 3/2\right)$$

with $n = n_x + n_y + n_z$

ii $|E_{n_x n_y n_z}\rangle = |E_{n_x}\rangle \otimes |E_{n_y}\rangle \otimes |E_{n_z}\rangle$

$$\langle x, y, z | E_{n_x n_y n_z}\rangle = \langle x | E_{n_x}\rangle \langle y | E_{n_y}\rangle \langle z | E_{n_z}\rangle$$

$$\psi_{n_x n_y n_z}(x,y,z) = \psi_{n_x}(x)\,\psi_{n_y}(y)\,\psi_{n_z}(z)$$

$$= \frac{e^{-\frac{1}{2\Delta^2}(x^2+y^2+z^2)}}{\sqrt{(\sqrt{\pi}\Delta)^3\, 2^n\, n_x!\, n_y!\, n_z!}}\, H_{n_x}\!\left(\frac{x}{\Delta}\right) H_{n_y}\!\left(\frac{y}{\Delta}\right) H_{n_z}\!\left(\frac{z}{\Delta}\right)$$

with $\Delta = \sqrt{\dfrac{\hbar}{m\omega}}$

Under parity $(x,y,z) \rightarrow (-x,-y,-z)$

$$H_{n_x}\!\left(\frac{x}{\Delta}\right) \rightarrow H_{n_x}\!\left(-\frac{x}{\Delta}\right) = (-1)^{n_x} H_{n_x}\!\left(\frac{x}{\Delta}\right) \text{, etc.}$$

$$\psi_{n_x n_y n_z}(x,y,z) \rightarrow \psi_{n_x n_y n_z}(-x,-y,-z)$$

$$= (-1)^{n_x+n_y+n_z}\, \psi_{n_x n_y n_z}(x,y,z)$$

$$\psi_{000}(r,\theta,\varphi) = \frac{1}{(\sqrt{\pi}\,\Delta)^{3/2}}\, e^{-\frac{1}{2}\frac{r^2}{\Delta^2}}$$

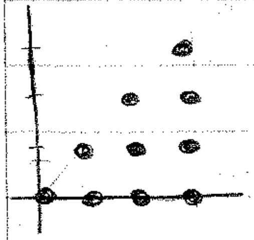
$$\psi_{100}(r,\theta,\varphi) = \frac{1}{\sqrt{(\sqrt{\pi}\,\Delta)^3\, 2}}\, e^{-\frac{1}{2}\frac{r^2}{\Delta^2}}\; \frac{2x}{\Delta}$$

$$= \sqrt{\frac{2}{\pi^{3/2}\,\Delta^5}}\; e^{-\frac{1}{2}\frac{r^2}{\Delta^2}}\; r \sin\theta \cos\varphi$$

$$\psi_{010}(r,\theta,\varphi) = \sqrt{\frac{2}{\pi^{3/2}\,\Delta^5}}\; e^{-\frac{1}{2}\frac{r^2}{\Delta^2}}\; r \sin\theta \sin\varphi$$

$$\psi_{001}(r,\theta,\varphi) = \sqrt{\frac{2}{\pi^{3/2}\,\Delta^5}}\; e^{-\frac{1}{2}\frac{r^2}{\Delta^2}}\; r \cos\theta$$

The degeneracy of the n^{th} level is



$$\sum_{n_x=0}^{\infty} \sum_{n_y=0}^{\infty} \sum_{n_z=0}^{\infty} \delta_{n,\, n_x+n_y+n_z}$$

$$= \sum_{n_x=0}^{n} \sum_{n_y=0}^{n-n_x} 1 = \sum_{n_x=0}^{n} (n-n_x+1)$$

$$= \sum_{m=0}^{n} (m+1) = \sum_{m=1}^{n+1} m = \frac{1}{2}(n+1)(n+2)$$

norm

S 10.3.2 $|3,3,4;S\rangle = N\big(|n_1=3, n_2=3, n_3=4\rangle$

$+ |n_1=3, n_2=4, n_3=3\rangle + |n_1=4, n_2=3, n_3=3\rangle\big)$

Clearly $N = \frac{1}{\sqrt{3}}$

S 10.3.3

i. $3^3 = 27$

ii. $\underset{(abc)}{1} + \underset{(abb)}{6} + \underset{(aaa)}{3} = 10$

iii. $\underset{(abc)}{1}$